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Abstract— (ε, δ)- DP-DIM Differentially Private Dense Interval Mining Window Sensitivity $\Delta f = 1$ DP DP Dense Interval Threshold Stability Sparse Vector Technique Privacy Budget Composition Apriori (i) $\varepsilon = 1.0$ F1 0.60 (ii) SVT (iii) (iv) $k = 500$ 77.5%

Index Terms— , , , , ,

I.

[1], [2] [3]

GDPR CCPA Differential Privacy, DP [5], [6] 1 [7]–[9]

1) 1 1 W 1
2) /
3) Apriori

- $\Delta_W = 1$ 1
- DP DP-DIM 1
- 3
- Sparse Vector Technique 2
- Apriori
-

II III IV DP-DIM V VI

II.

A.

Dwork [5] (ε, δ)- [6] [10], [11] Sparse Vector Technique (SVT) [12]–[14] Rényi [15] [16]

B.

Zeng [7] truncation FP-tree Li [8] PrivBasis Bhaskar [9] top-k

TABLE I

	DP		
Zeng [7]	×	✓	×
Li [8]	×	✓	×
Chan [18]	✓	✓	×
Kellaris [20]	✓	✓	×
Suzuki [3]	✓	×	✓
DP-DIM	✓	✓	✓

Su [17] Lee Clifton [4]

C.

[4] [3] Apriori W θ θ

D.

Chan [18] $O(\log^{1.5} T)$ Dwork [19] continual observation Kellaris [20] w-event Fan Xiong [21]

E.

I Apriori

III.

A.

1 ( ). $\mathcal{D}$ = $\{T_0, T_1, \ldots, T_{n-1}\}$ T_t $\mathcal{I}$ t

2 ( ). $X \subseteq \mathcal{I}$ l $C_W(X, l)$

$$C_W(X, l) = |\{t \mid l \leq t \leq l + W,\, X \subseteq T_t\}|$$

W

3 ([3]). X $C_W(X, l)$ $\geq$ θ l $[s, e]$ θ

B.

1 T_t 1

4 ( ). 2 $\mathcal{D}$ $\mathcal{D}'$ $\mathcal{D}$ $\sim$ $\mathcal{D}'$ t^* T_{t^*} $\neq$ T'_{t^*} t $\neq$ t^* T_t $=$ T'_t 1 , bounded adjacency

5 ((ε,δ)- [5], [6]). $\mathcal{M}$: $\mathcal{X}$ $\to$ $\mathcal{R}$ (ε,δ)- $\mathcal{D}\sim\mathcal{D}'$ 4 $S\subseteq\mathcal{R}$

$$\Pr[\mathcal{M}(\mathcal{D})\in S]\le e^{\varepsilon}\cdot\Pr[\mathcal{M}(\mathcal{D}')\in S]+\delta$$

δ $=$ 0 ε-DP ε $>$ 0 privacy loss ε $\in$ $[0.1,10]$ ε $\le$ 1 ε $\ge$ 5 [6]

6 ([5]). $f:\mathcal{X}\to\mathbb{R}^d$ L_p- $\Delta_p f$

$$\Delta_p f=\max_{\mathcal{D}\sim\mathcal{D}'}\|f(\mathcal{D})-f(\mathcal{D}')\|_p$$

L_1 $\Delta_1 f$ L_2 $\Delta_2 f$

7 ([5]). f : $\mathcal{X}$ $\to$ $\mathbb{R}$ $\mathcal{M}(\mathcal{D})$ $=$ $f(\mathcal{D})$ $+$ $\mathrm{Lap}(\Delta_1 f/\varepsilon)$ $\mathrm{Lap}(b)$ b $p(x)$ $=$ $\frac{1}{2b}e^{-|x|/b}$ ε-DP

8 ([6]). f : $\mathcal{X}$ $\to$ $\mathbb{R}$ $\mathcal{M}(\mathcal{D})$ $=$ $f(\mathcal{D})+\mathcal{N}(0,\sigma^2)$ σ $\ge$ $\Delta_2 f\cdot\sqrt{2\ln(1.25/\delta)}/\varepsilon$ ε $\in$ $(0,1)$ (ε,δ)-DP

C.

1 ([6]). $\mathcal{M}_1,\dots,\mathcal{M}_k$ (ε_i,δ_i)-DP $(\mathcal{M}_1,\dots,\mathcal{M}_k)$ $(\sum_i\varepsilon_i,\sum_i\delta_i)$-DP

2 ([10], [11]). k ε_0-DP δ' $>$ 0 $(\varepsilon',k\delta+\delta')$-DP

$$\varepsilon'=\varepsilon_0\sqrt{2k\ln(1/\delta')}+k\varepsilon_0(e^{\varepsilon_0}-1)$$

ε_0 $\le$ 1 2 $O(k\varepsilon_0^2)$ ε' $\approx$ $\varepsilon_0\sqrt{2k\ln(1/\delta')}$ k $O(\sqrt{k})$ $O(k)$

IV. DP-DIM

DP-DIM D-P SVT Apriori

A.

DP

9 (Window Sensitivity). X l $f_{X,l}(\mathcal{D})$ $=$ $C_W(X,l;\mathcal{D})$ L_1- Δ_W

$$\Delta_W=\max_{\mathcal{D}\sim\mathcal{D}'}|C_W(X,l;\mathcal{D})-C_W(X,l;\mathcal{D}')|$$

$\mathcal{D}\sim\mathcal{D}'$ 4

1 ( ). X W l $\Delta_W=1$

Proof. $\mathcal{D}\sim\mathcal{D}'$ 4 t^* $T_{t^*}\neq T'_{t^*}$ $t\neq t^*$ $T_t=T'_t$ l

$|C_W(X,l;\mathcal{D})-C_W(X,l;\mathcal{D}')|=|\mathbf{1}[X\subseteq T_{t^*}]-\mathbf{1}[X\subseteq T'_{t^*}]|\cdot\mathbf{1}[l\le t^*\le l+W]$

Algorithm 1 DP-DIM:

Require: $\mathrm{VI}^{\mathbf{t}}$ W θ ε
Ensure: DP $\mathcal{I}$

```
L ←
ε_q ← ε/|L|                                  {        }
b ← Δ_W/ε_q                                  {        }
I ← ∅
in_dense ← false
for        l ∈ L do
    c ← C_W(X, l)                            {    }
    c̃ ← c + Lap(b)                           {    }
    if c̃ ≥ θ then
        if ¬in_dense then
            start ← l
        end if
        end ← l
        in_dense ← true
    else
        if in_dense then
            I ← I ∪ {(start, end)}
        end if
        in_dense ← false
    end if
end for
if in_dense then
    I ← I ∪ {(start, end)}
end if
return I
```

$\mathbf{1}[\cdot]$ t^* $[l,l+W]$ 0 $\{-1,0,1\}$ X T_{t^*} $X\not\subseteq T'_{t^*}$ ±1 $\Delta_W=1$

$f_{X,l}$ 1 $\mathbf{f}_X(\mathcal{D})$ $=$ $(C_W(X,0),\dots,C_W(X,n-W-1))$ L_1 t^* $\min(W+1,n-W)$ $\Delta_1\mathbf{f}_X$ $=$ $\min(W+1,n-W)$ $\Delta_W=1$ □

B. DP

10 (DP DP Dense Interval). $\tilde{C}_W(X,l)$ $=$ $C_W(X,l)$ $+$ η η $\tilde{C}_W(X,l)$ $\ge$ θ l DP

2 (DP-DIM). 1 ε-

Proof. l ε_q $=$ $\varepsilon/|L|$, Δ_W $=$ 1 $|L|$ $|L|\cdot\varepsilon_q=\varepsilon$-DP □

C.

/

11 (Threshold Stability). $\tilde{c}$ θ γ

$$\gamma=\frac{\Delta_W}{\varepsilon_q}\cdot\ln\frac{1}{1-p}$$

p $|\tilde{c}-\theta|\ge\gamma$ p

3 ( ). c $|c$ $-$ $\theta|$ $\ge$ $\tilde{c}$ p

Algorithm 2 SVT-DIM: SVT DP

Require: θ ε $c_{\max}$
Ensure: above-threshold

```
ε_1 ← ε/2, ε_2 ← ε/2
θ̃ ← θ + Lap(Δ_W/ε_1)                    {    }
count ← 0
for    l do
    ν_l ← Lap(2c_max · Δ_W/ε_2)         {    }
    if C_W(X, l) + ν_l ≥ θ̃ then
        : l  above-threshold
        count ← count + 1
        if count ≥ c_max then
            
        end if
    else
        : l  below-threshold ⊥
    end if
end for
```

Proof. Lap(b) $b = \Delta_W/\varepsilon_q$ $\Pr[|\eta| < b \cdot \ln(1/(1-p))] = p$ $|c-\theta| \geq \gamma = b \cdot \ln(1/(1-p))$ $|\eta| < \gamma$ $\operatorname{sign}(\tilde{c}-\theta) = \operatorname{sign}(c-\theta)$ p □

D. Sparse Vector Technique

Sparse Vector Technique (SVT) [12], [14] SVT below-threshold

SVT ε above-threshold $c_{\max}$ $|L|$ [14] $|L|$ SVT SVT above/below

E.

Apriori DP-DIM

12 ( ). Apriori K ε

1) k $k = 1, \ldots, K$ $\varepsilon_k = \varepsilon/K$
2) k m_k ε_k/m_k

4 ( ). 12 DP-DIM ε-

Proof. 1 ε_k/m_k-DP k m_k k $m_k \cdot (\varepsilon_k/m_k) = \varepsilon_k = \varepsilon/K$-DP K $K \cdot (\varepsilon/K) = \varepsilon$-DP □

2

1 ( ). $M = \sum_{k=1}^{K} m_k$ · $|L_k|$ $|L_k|$ k ε_0 $= \varepsilon/M$ ε_0

$$\varepsilon'_{\text{adv}} = \varepsilon_0\sqrt{2M\ln(1/\delta')} + M\varepsilon_0(e^{\varepsilon_0} - 1)$$

$\varepsilon_0 \ll 1$ $\varepsilon'_{\text{adv}} \approx \varepsilon_0\sqrt{2M\ln(1/\delta')} = O(\sqrt{M})$ $\varepsilon_{\text{seq}} = M\varepsilon_0 = O(M)$ $O(\sqrt{M})$

TABLE II
E1: ε Jaccard F1 $W = 20$, $\theta = 5$, 10

ε	Jaccard ( )	Jaccard ( )	F1 ( )	F1 ( )
0.1	0.429	0.031	0.598	0.030
0.5	0.430	0.031	0.599	0.029
1.0	0.431	0.030	0.600	0.029
2.0	0.433	0.029	0.602	0.028
5.0	0.439	0.030	0.608	0.029
10.0	0.449	0.029	0.617	0.027

TABLE III
E2: $\varepsilon = 2.0$, 10

	Jaccard ( )	Jaccard ( )	F1 ( )	(ms)
Laplace	0.348	0.083	0.516	0.31
Gaussian ($\delta = 10^{-5}$)	0.311	0.049	0.475	0.37
SVT ($c_{\max} = 50$)	0.233	0.089	0.378	0.11
DP	0.461	0.025	0.631	0.28

V.

A.

DP-DIM n n = 300 100–2000 $|\mathcal{I}|$ = 20 $[s_d, e_d]$ X_d 0.85 0.05 $W = 20$ $\theta = 5$ $K = 3$ Jaccard F1 precision/recall 10 42–51 DP ground truth

B. E1:

ε $\{0.1, 0.5, 1.0, 2.0, 5.0, 10.0\}$ DP Jaccard F1 $c_{\max} \ll$ II

ε Jaccard F1 $\varepsilon = 10.0$ Jaccard 0.449 F1 0.617 $\varepsilon = 1.0$ F1 0.60

DP Jaccard 0.461 ε = 1.0 DP $(0.461 - 0.431)/0.461 = 6.5\%$ ε = 0.1 6.9% 1 D-P

C. E2:

$\delta = 10^{-5}$ SVT $c_{\max} = 50$ 3 ε = 2.0 III

Jaccard 0.348 (ε, δ)-DP SVT $|L|$ DP $\varepsilon = 2.0$ 24.5%

D. E3:

n $\{100, 200, 500, 1000, 2000\}$ IV n $R^2 = 0.999$ $n = 2000$ 51 ms DP

E. E4:

ε p γ V 3 γ p $\varepsilon = 2.0$ $p = 0.95$ 1.50 2 95% $\varepsilon = 5.0$ $\gamma = 0.60$ $p = 0.95$ $\theta \geq 5$ 0 θ $|c-\theta| \geq 2$ $\varepsilon \geq 1.0$

TABLE IV
E3: $\varepsilon = 1.0,\ 3$

n	(ms,)	(ms,)
100	2.2	0.1
200	4.5	0.0
500	12.2	0.2
1000	24.9	0.9
2000	51.2	1.1

TABLE V
E4: $\gamma\ \Delta_W = 1$,

ε	$p = 0.80$	$p = 0.90$	$p = 0.95$	$p = 0.99$
0.5	3.22	4.61	5.99	9.21
1.0	1.61	2.30	3.00	4.61
2.0	0.80	1.15	1.50	2.30
5.0	0.32	0.46	0.60	0.92

F. E5:

ε VI $\varepsilon_0 = 0.01$ $k = 10$ $\sqrt{k \ln(1/\delta')}$ $k \geq 50$ $k = 500$ 77.5%

G. E6:

W θ DP VII $\varepsilon = 1.0$ $W \in \{10, 20, 40\}$ $\theta \in \{3, 5, 10\}$ θ DP DP θ W SNR $\theta/\gamma \geq 2$ $\theta \geq 2\gamma$

TABLE VI
E5: vs $\varepsilon_0 = 0.01,\ \delta' = 10^{-5}$

k	ε	ε	
10	0.10	0.15	−52.7%
50	0.50	0.34	31.1%
100	1.00	0.49	51.0%
500	5.00	1.12	77.5%

TABLE VII
E6: $\varepsilon = 1.0,\ n = 300,\ 10$ F1

		θ		
W	DP F1	$\theta = 3$	$\theta = 5$	$\theta = 10$
10	0.583	0.541	0.572	0.580
20	0.631	0.558	0.600	0.624
40	0.672	0.587	0.639	0.667

VI.

(ε, δ)- DP-DIM (1) $\Delta_W = 1$ (2) (3) DP (4) (5) SVT (6) Apriori

- $\varepsilon = 1.0$ F1 0.60 DP 6.5%
- SVT DP
- DP
- $k = 500$ 77.5%
- θ W DP $\theta \geq 2\gamma$

A.

1 1 $\min(B, W{+}1)$ [6] ε B Dunnhumby, Online Retail D-P Rényi RDP [15] zCDP [16] Apriori DP Encoding

DP 1 B Δ_W [22] LDP LDP OUE Unary

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